

BASIC NETWORK SCANNING WITH NMAP

Professional Network Security Assessment Report

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Environment: Windows / Nmap 7.991

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Assessment Target: 10.27.217.62

Authorized local/lab assessment only

1. Objective

The objective of this assessment was to perform a basic network scan using Nmap, identify open TCP ports and running services, detect service versions, identify the operating system, and document security considerations and recommendations.

2. Scope and Environment

Scanner: Windows system

Tool: Nmap 7.991

Target: 10.27.217.62

Assessment type: Authorized local/lab network scan

The target address used below is the address contained in the supplied Nmap scan evidence. It should only be submitted if the target was owned by the tester or explicitly authorized for scanning.

3. Methodology

The assessment used a staged Nmap workflow: basic host/port discovery, service and version detection, operating-system detection, and a combined service/OS scan.

Commands used:

```
nmap 10.27.217.62
```

```
nmap -sV 10.27.217.62
```

```
nmap -O 10.27.217.62
```

```
nmap -sV -O 10.27.217.62
```

4. Findings

Port	State	Service / Version	Security Consideration
135/tcp	Open	MSRPC — Microsoft Windows RPC	Medium: network RPC exposure increases attack surface if unnecessarily accessible.
139/tcp	Open	NetBIOS-SSN — Microsoft Windows NetBIOS	Medium: legacy Windows networking exposure should be restricted where unnecessary.
445/tcp	Open	Microsoft-DS / SMB	Medium–High: SMB is a significant Windows network

			attack surface when unnecessarily exposed.
5000/tcp	Open	HTTP — Werkzeug httpd 3.1.9 / Python 3.11.9	Medium: web application exposure requires configuration, patching and access-control review.

5. Service Analysis

Port 135 — MSRPC

Microsoft RPC supports communication between Windows components and network services.

Recommendation: Restrict RPC access with firewall rules, allow trusted hosts only, and maintain Windows security updates.

Port 139 — NetBIOS

NetBIOS Session Service supports legacy Windows network communication.

Recommendation: Disable NetBIOS where not required and restrict access using firewall controls.

Port 445 — SMB

SMB supports Windows file/printer sharing and can represent a significant network attack surface when unnecessarily exposed.

Recommendation: Restrict SMB to trusted networks, keep Windows patched, use strong authentication, and disable unnecessary SMB features.

Port 5000 — HTTP

Nmap identified Werkzeug httpd 3.1.9 with Python 3.11.9. Recommendation: Verify the service is required, keep application dependencies updated, avoid exposing development servers to untrusted networks, and review authentication and access controls.

6. OS Detection

Nmap identified the target as a general-purpose Microsoft Windows 11 system, with OS details reported as Windows 11 24H2–25H2. Network distance was reported as 0 hops.

7. Security Recommendations

- Keep Windows fully patched.
- Review all exposed network services.
- Restrict SMB and RPC access to trusted systems.
- Disable unnecessary legacy services.

- Review Windows Firewall rules.
- Verify whether the HTTP service on TCP/5000 is required.
- Keep applications and dependencies updated.
- Regularly review open ports and running services.

8. Limitations

Nmap service and OS detection provide identification information but do not, by themselves, prove that a specific vulnerability exists. This assessment should therefore be considered a network enumeration and basic security-analysis exercise rather than a complete vulnerability assessment.

9. Ethical Considerations

Network scanning must only be performed against systems owned by the tester or systems for which explicit permission has been provided. Public, production, or third-party systems should not be scanned without authorization.

10. Evidence

Supporting evidence consists of the Nmap command outputs and screenshots collected during the assessment.

- 01_nmap_version.png
- 02_ipconfig.png
- 03_ping_test.png
- 04_basic_scan.png
- 05_service_version.png
- 06_os_detection.png
- 07_final_scan.png